

• Exercise 6.3

1. Find the adjoint of each of the following matrices. Also, find the inverse whenever possible.

a) $\begin{pmatrix} 3 & -1 \\ 2 & 4 \end{pmatrix}$

→ let, $A = \begin{pmatrix} 3 & -1 \\ 2 & 4 \end{pmatrix}$

$A_{11} = 4$

$A_{12} = -2$

$A_{21} = -(-1) = 1$

$A_{22} = 3$

$\text{Adj. } A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}^T = \begin{pmatrix} 4 & -2 \\ 1 & 3 \end{pmatrix}^T = \begin{pmatrix} 4 & 1 \\ -2 & 3 \end{pmatrix}$

Now,

$|A| = \begin{vmatrix} 3 & -1 \\ 2 & 4 \end{vmatrix} = 12 - (-2) = 14 \neq 0$

$A^{-1} = \frac{1}{|A|} \text{Adj. of } A$

$= \frac{1}{14} \begin{pmatrix} 4 & 1 \\ -2 & 3 \end{pmatrix}$

b) $\begin{pmatrix} 6 & -2 \\ 1 & -3 \end{pmatrix}$

→ let, $A = \begin{pmatrix} 6 & -2 \\ 1 & -3 \end{pmatrix}$

$A_{11} = -3$

$A_{12} = -1$

$A_{21} = -(-2) = 2$

$A_{22} = 6$

$$\text{Adj. } A = \begin{pmatrix} -3 & -1 \\ 2 & 6 \end{pmatrix}^T = \begin{pmatrix} -3 & 2 \\ -1 & 6 \end{pmatrix}$$

$$|A| = \begin{vmatrix} 6 & -2 \\ 1 & -3 \end{vmatrix} = -18 - (-2) = -18 + 2 = -16$$

$\therefore |A| \neq 0$, so inverse exist

Now,

$$A^{-1} = \frac{1}{|A|} \text{Adj. } A$$

$$= \frac{1}{-16} \begin{pmatrix} -3 & 2 \\ -1 & 6 \end{pmatrix}$$

$$= \frac{1}{16} \begin{pmatrix} 3 & -2 \\ 1 & -6 \end{pmatrix}$$

$$\therefore A^{-1} = \frac{1}{16} \begin{pmatrix} 3 & -2 \\ 1 & -6 \end{pmatrix} \checkmark$$

$$c) \begin{pmatrix} 1 & 2 & -1 \\ 2 & 0 & 1 \\ 0 & 3 & -1 \end{pmatrix}$$

$$\rightarrow \text{let, } A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 0 & 1 \\ 0 & 3 & -1 \end{pmatrix}$$

$$A_{11} = \begin{vmatrix} 0 & 1 \\ 3 & -1 \end{vmatrix} = 0 - 3 = -3$$

$$A_{12} = - \begin{vmatrix} 2 & 1 \\ 0 & -1 \end{vmatrix} = -(-2) = 2$$

$$A_{13} = \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} = 6 - 0 = 6$$

$$A_{21} = - \begin{vmatrix} 2 & -1 \\ 3 & -1 \end{vmatrix} = -(-2 + 3) = -1$$

$$A_{22} = \begin{vmatrix} 1 & -1 \\ 0 & -1 \end{vmatrix} = -1 + 0 = -1$$

$$A_{23} = - \begin{vmatrix} 1 & 2 \\ 0 & 3 \end{vmatrix} = -(3 - 0) = -3$$

$$A_{31} = \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} = 2 + 0 = 2$$

$$A_{32} = - \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = -(1 + 2) = -3$$

$$A_{33} = \begin{vmatrix} 1 & 2 \\ 2 & 0 \end{vmatrix} = 0 - 4 = -4$$

For determinant,
expanding along R_1 ,

$$\begin{aligned} & 1 \begin{vmatrix} 0 & 1 \\ 3 & -1 \end{vmatrix} - 2 \begin{vmatrix} 2 & 1 \\ 0 & -1 \end{vmatrix} + (-1) \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} \\ &= 1(0 - 3) - 2(-2 - 0) - 1(6 - 0) \\ &= -3 + 4 - 6 \\ &= -5 \neq 0, \therefore A^{-1} \text{ exists,} \end{aligned}$$

Now,

$$A^{-1} = \frac{1}{|A|} \text{Adj. } A$$

$$= \frac{1}{-5} \begin{pmatrix} -3 & 2 & 6 \\ -1 & -1 & -3 \\ 2 & -3 & -4 \end{pmatrix}^T$$

$$= \frac{1}{-5} \begin{pmatrix} -3 & -1 & 2 \\ 2 & -1 & -3 \\ 6 & -3 & -4 \end{pmatrix}$$

$$= \begin{pmatrix} 3/5 & 1/5 & -2/5 \\ -2/5 & 1/5 & 3/5 \\ -6/5 & 3/5 & 4/5 \end{pmatrix}$$

Ans.

$$d) \begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & -2 \\ -3 & -1 & 3 \end{pmatrix}$$

$$\rightarrow \text{let, } A = \begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & -2 \\ -3 & -1 & 3 \end{pmatrix}$$

$$A_{11} = \begin{vmatrix} 1 & -2 \\ -1 & 3 \end{vmatrix} = 3 - 2 = 1$$

$$A_{12} = - \begin{vmatrix} 0 & -2 \\ -3 & 3 \end{vmatrix} = -(0 - 6) = +6$$

$$A_{13} = \begin{vmatrix} 0 & 1 \\ -3 & -1 \end{vmatrix} = 0 + 3 = 3$$

$$A_{21} = - \begin{vmatrix} 3 & 2 \\ -1 & 3 \end{vmatrix} = -(9 + 2) = -11$$

$$A_{22} = \begin{vmatrix} 1 & 2 \\ -3 & 3 \end{vmatrix} = 3 + 6 = 9$$

$$A_{23} = - \begin{vmatrix} 1 & 3 \\ -3 & -1 \end{vmatrix} = -(-1 + 9) = -8$$

$$A_{31} = \begin{vmatrix} 3 & 2 \\ 1 & -2 \end{vmatrix} = -6 - 2 = -8$$

$$A_{32} = - \begin{vmatrix} 1 & 2 \\ 0 & -2 \end{vmatrix} = -(-2 - 0) = +2$$

$$A_{33} = \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 1 - 0 = 1$$

$$\text{Adj. } A = \begin{pmatrix} 1 & 6 & 3 \\ -11 & 9 & -8 \\ -8 & 2 & 1 \end{pmatrix}^T$$

$$= \begin{pmatrix} 1 & -11 & -8 \\ 6 & 9 & 2 \\ 3 & -8 & 1 \end{pmatrix}$$

Now,

$$|A| = \begin{vmatrix} 1 & 3 & 2 \\ 0 & 1 & -2 \\ -3 & -1 & 3 \end{vmatrix}$$

Expanding along R_1 ,

$$= 1 \begin{vmatrix} 1 & -2 \\ -1 & 3 \end{vmatrix} - 3 \begin{vmatrix} 0 & -2 \\ -3 & 3 \end{vmatrix} + 2 \begin{vmatrix} 0 & 1 \\ -3 & -1 \end{vmatrix}$$

$$= 1(3 - 2) - 3(0 - 6) + 2(0 + 3)$$

$$= 1 + 18 + 6$$

$$= 25 \neq 0 \therefore A^{-1} \text{ exists,}$$

$$A^{-1} = \frac{1}{|A|} \text{Adj. } A$$

$$= \frac{1}{25} \begin{pmatrix} 1 & -11 & -8 \\ 6 & 9 & 2 \\ 3 & -8 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1/25 & -11/25 & -8/25 \\ 6/25 & 9/25 & 2/25 \\ 3/25 & -8/25 & 1/25 \end{pmatrix}$$

$$\therefore A^{-1} = \begin{pmatrix} 1/25 & -11/25 & -8/25 \\ 6/25 & 9/25 & 2/25 \\ 3/25 & -8/25 & 1/25 \end{pmatrix} \quad \checkmark$$

② If $\begin{pmatrix} 5 & 2n+3 \\ -1 & 3n-1 \end{pmatrix}$ is not invertible, find the value of n .

→ Soln,

$\begin{pmatrix} 5 & 2n+3 \\ -1 & 3n-1 \end{pmatrix}$ is not invertible, So

$$\begin{vmatrix} 5 & 2n+3 \\ -1 & 3n-1 \end{vmatrix} = 0$$

$$\text{or, } 5(3n-1) - (-1)(2n+3) = 0$$

$$\text{or, } 15n - 5 + 2n + 3 = 0$$

$$\text{or, } 17n - 2 = 0$$

$$\text{or, } 17n = 2$$

$$\text{or, } n = \frac{2}{17}$$

$$\therefore n = \frac{2}{17} \text{ ✓}$$

3. ~~Pro~~

a) Prove that the matrices $\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$ and $\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$

are inverses of each other.

$$\rightarrow \text{let, } A = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$$

$$A \cdot B = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 4-3 & 2-2 \\ -6+6 & -3+4 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$B \cdot A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 4-3 & -2+2 \\ 6-6 & -3+4 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Since, $A \cdot B = B \cdot A = I$ A and B are inverses of each other.

Proved

b) Show that the matrices $\begin{pmatrix} -3 & -7 \\ 1 & 2 \end{pmatrix}$ and $\begin{pmatrix} 2 & 7 \\ -1 & -3 \end{pmatrix}$ are inverses of each other.

→ Soln,

$$\text{let } A = \begin{pmatrix} -3 & -7 \\ 1 & 2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 2 & 7 \\ -1 & -3 \end{pmatrix}$$

$$\begin{aligned} A \cdot B &= \begin{pmatrix} -3 & -7 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 & 7 \\ -1 & -3 \end{pmatrix} \\ &= \begin{pmatrix} -6+7 & -21+21 \\ 2-2 & 7-6 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \end{aligned}$$

$$\begin{aligned} B \cdot A &= \begin{pmatrix} 2 & 7 \\ -1 & -3 \end{pmatrix} \cdot \begin{pmatrix} -3 & -7 \\ 1 & 2 \end{pmatrix} \\ &= \begin{pmatrix} -6+7 & -14+14 \\ 3-3 & 7-6 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \end{aligned}$$

Since, $A \cdot B = B \cdot A = I$, So, A and B are inverses of each other.

4.

a) If $A = \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix}$, Verify that $A \cdot (\text{adj. } A) = (\text{adj. } A) \cdot A = |A| \cdot I$

→ Soln,

$$A_{11} = 4$$

$$A_{12} = -(-3) = 3$$

$$A_{21} = -2$$

$$A_{22} = 1$$

$$\text{Adj. } A = \begin{pmatrix} 4 & 3 \\ -2 & 1 \end{pmatrix}^T = \begin{pmatrix} 4 & -2 \\ 3 & 1 \end{pmatrix}$$

$$|A| = \begin{vmatrix} 1 & 2 \\ -3 & 4 \end{vmatrix} = 4 + 6 = 10$$

Now,

$$\begin{aligned} A \cdot (\text{adj. } A) &= \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix} \cdot \begin{pmatrix} 4 & -2 \\ 3 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 4+6 & -2+2 \\ -12+12 & 6+4 \end{pmatrix} \\ &= 10 \end{aligned}$$

$$\begin{aligned} (\text{adj. } A) \cdot A &= \begin{pmatrix} 4 & -2 \\ 3 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 4+6 & 8-8 \\ 3-3 & 6+4 \end{pmatrix} \\ &= \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix} \end{aligned}$$

$$|A| \cdot I = 10 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix}$$

$$\therefore A \cdot (\text{adj. } A) = (\text{adj. } A) \cdot A = |A| \cdot I = \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix} \quad \text{✓}$$

b) If $A = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 3 & -2 \\ -2 & 1 & -3 \end{pmatrix}$, verify that

$$A \cdot (\text{adj. } A) = (\text{adj. } A) \cdot A = |A| \cdot I$$

⇒ Soln,

$$A_{11} = \begin{vmatrix} 3 & -2 \\ 1 & -3 \end{vmatrix} = -9 + 2 = -7$$

$$A_{12} = - \begin{vmatrix} 0 & -2 \\ -2 & -3 \end{vmatrix} = -(0 - 4) = 4$$

$$A_{13} = \begin{vmatrix} 0 & 3 \\ -2 & 1 \end{vmatrix} = (0 + 6) = 6$$

$$A_{21} = - \begin{vmatrix} -1 & 1 \\ 1 & -3 \end{vmatrix} = -(3 - 1) = -2$$

$$A_{22} = \begin{vmatrix} 2 & 1 \\ -2 & -3 \end{vmatrix} = (-6 + 2) = -4$$

$$A_{23} = - \begin{vmatrix} 2 & -1 \\ -2 & 1 \end{vmatrix} = -(2 - 2) = 0$$

$$A_{31} = \begin{vmatrix} -1 & 1 \\ 3 & -2 \end{vmatrix} = (2 - 3) = -1$$

$$A_{32} = - \begin{vmatrix} 2 & 1 \\ 0 & -2 \end{vmatrix} = -(-4 - 0) = 4$$

$$A_{33} = \begin{vmatrix} 2 & -1 \\ 0 & 3 \end{vmatrix} = (6 - 0) = 6$$

$$\text{Adj. } A = \begin{pmatrix} -7 & 4 & 6 \\ -2 & -4 & 0 \\ -1 & 4 & 6 \end{pmatrix}^T$$

$$= \begin{pmatrix} -7 & -2 & -1 \\ 4 & -4 & 4 \\ 6 & 0 & 6 \end{pmatrix}$$

Now,

$$A \cdot (\text{Adj. } A) = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 3 & -2 \\ -2 & 1 & -3 \end{pmatrix} \cdot \begin{pmatrix} -7 & -2 & -1 \\ 4 & -4 & 4 \\ 6 & 0 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} -14 + 4 + 6 & -4 + 4 + 0 & -2 - 4 + 6 \\ 0 + 12 - 12 & 0 - 12 - 0 & 0 + 12 - 12 \\ 14 + 4 - 18 & 4 - 4 + 0 & 2 + 4 - 18 \end{pmatrix}$$

$$= \begin{pmatrix} -12 & 0 & 0 \\ 0 & -12 & 0 \\ 0 & 0 & -12 \end{pmatrix}$$

$$(Adj \cdot A) \cdot A = \begin{pmatrix} -7 & -2 & -1 \\ 2 & -4 & 4 \\ 6 & 0 & 6 \end{pmatrix} \cdot \begin{pmatrix} 2 & -1 & 1 \\ 0 & 3 & -2 \\ -2 & 1 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} -14 - 0 + 2 & 7 - 6 - 1 & -7 + 4 + 3 \\ 8 - 0 - 8 & -2 - 12 + 4 & 2 + 8 - 12 \\ 12 - 0 - 12 & -6 + 0 + 6 & 6 - 0 - 18 \end{pmatrix}$$

$$= \begin{pmatrix} -12 & 0 & 0 \\ 0 & -12 & 0 \\ 0 & 0 & -12 \end{pmatrix}$$

$$|A| = \begin{vmatrix} 2 & -1 & 1 \\ 0 & 3 & -2 \\ -2 & 1 & -3 \end{vmatrix}$$

expanding along R_1 .

$$|A| = 2 \begin{vmatrix} 3 & -2 \\ 1 & -3 \end{vmatrix} + 1 \begin{vmatrix} 0 & -2 \\ -2 & -3 \end{vmatrix} + 1 \begin{vmatrix} 0 & 3 \\ -2 & 1 \end{vmatrix}$$

$$= 2(-9 + 2) + 1(0 - 4) + 1(0 + 6)$$

$$= -14 - 4 + 6$$

$$= -12$$

$$|A| \cdot I = 12 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{pmatrix}$$

$$\therefore A \cdot (Adj \cdot A) = (Adj \cdot A) \cdot A = |A| \cdot I = \begin{pmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{pmatrix}$$

Q. If $A = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} -3 & -2 \\ 1 & 2 \end{pmatrix}$, verify

that $(AB)^{-1} = B^{-1} \cdot A^{-1}$

→ Soln,

$$\begin{aligned} A \cdot B &= \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} -3 & -2 \\ 1 & 2 \end{pmatrix} \\ &= \begin{pmatrix} -6+1 & -4+2 \\ 3+1 & 2+2 \end{pmatrix} \\ &= \begin{pmatrix} -5 & -2 \\ 4 & 4 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} AB_{11} &= -5 & AB_{12} &= -2 \\ AB_{21} &= 4 & AB_{22} &= 4 \end{aligned}$$

$$\text{Adj. } AB = \begin{pmatrix} 4 & -4 \\ 2 & -5 \end{pmatrix}^T = \begin{pmatrix} 4 & 2 \\ -4 & -5 \end{pmatrix}$$

$$\begin{aligned} |AB| &= \begin{vmatrix} -5 & -2 \\ 4 & 4 \end{vmatrix} \\ &= -20 + 8 \\ &= -12 \end{aligned}$$

Now,

$$AB^{-1} = \frac{1}{|AB|} \text{Adj. of } AB$$

$$\begin{aligned} &= \frac{1}{-12} \begin{pmatrix} 4 & 2 \\ -4 & -5 \end{pmatrix} \\ &= \begin{pmatrix} -\frac{1}{3} & -\frac{1}{6} \\ \frac{1}{3} & \frac{5}{12} \end{pmatrix} \end{aligned}$$

Similarly,

$$B_{11} = 2$$

$$B_{21} = -(-2) = 2$$

$$B_{12} = -1$$

$$B_{22} = -3$$

$$\text{Adj. } B = \begin{pmatrix} 2 & -1 \\ 2 & -3 \end{pmatrix}^T = \begin{pmatrix} 2 & 2 \\ -1 & -3 \end{pmatrix}$$

$$|B| = \begin{vmatrix} -3 & -2 \\ 1 & 2 \end{vmatrix} = -6 + 2 = -4$$

$$B^{-1} = \frac{\text{Adj. } B}{|B|}$$
$$= \frac{1}{-4} \begin{pmatrix} 2 & 2 \\ -1 & -3 \end{pmatrix}$$

$$A_{11} = 1$$

$$A_{21} = -1$$

$$A_{12} = -(-1) = 1$$

$$A_{22} = 2$$

$$\text{Adj. } A = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}^T = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$$

$$|A| = \begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix} = 2 + 1 = 3$$

$$A^{-1} = \frac{1}{|A|} \text{Adj. } A$$

$$= \frac{1}{3} \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$$

Again,

$$B^{-1} \cdot A^{-1} = \frac{1}{-4} \begin{pmatrix} 2 & 2 \\ -1 & -3 \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$$
$$= \frac{1}{-12} \begin{pmatrix} 2+2 & -2+4 \\ -1-3 & 1-6 \end{pmatrix}$$

$$= \frac{1}{-12} \begin{pmatrix} 4 & 2 \\ -4 & -5 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{1}{3} & -\frac{1}{6} \\ \frac{1}{3} & \frac{5}{12} \end{pmatrix}$$

$$\therefore (AB)^{-1} = B^{-1} \cdot A^{-1} = \begin{pmatrix} -1/3 & -1/6 \\ 1/3 & 5/12 \end{pmatrix}$$

6. If $A = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 1 & -2 \\ 3 & 0 & -3 \end{pmatrix}$, verify that $A \cdot A^{-1} = I$

$$A^{-1} \cdot A = I.$$

→ Soln,

$$|A| = \begin{vmatrix} 2 & -1 & 1 \\ 0 & 1 & -2 \\ 3 & 0 & -3 \end{vmatrix}$$

expanding along C_1

$$= 2 \begin{vmatrix} 1 & -2 \\ 0 & -3 \end{vmatrix} - 0 + 3 \begin{vmatrix} -1 & 1 \\ 1 & -2 \end{vmatrix}$$

$$= 2(-3 - 0) + 3(2 - 1)$$

$$= -6 + 3$$

$$= -3$$

$$A_{11} = \begin{vmatrix} 1 & -2 \\ 0 & -3 \end{vmatrix} = (-3 - 0) = -3$$

$$A_{12} = - \begin{vmatrix} 0 & -2 \\ 3 & -3 \end{vmatrix} = -(0 + 6) = -6$$

$$A_{13} = \begin{vmatrix} 0 & 1 \\ 3 & 0 \end{vmatrix} = (0-3) = -3$$

$$A_{21} = - \begin{vmatrix} -1 & 1 \\ 0 & -3 \end{vmatrix} = -(3-0) = -3$$

$$A_{22} = \begin{vmatrix} 2 & 1 \\ 3 & -3 \end{vmatrix} = (-6-3) = -9$$

$$A_{23} = - \begin{vmatrix} 2 & -1 \\ 3 & 0 \end{vmatrix} = -(0+3) = -3$$

$$A_{31} = \begin{vmatrix} -1 & 1 \\ 1 & -2 \end{vmatrix} = (2-1) = 1$$

$$A_{32} = - \begin{vmatrix} 2 & 1 \\ 0 & -2 \end{vmatrix} = -(-4-0) = 4$$

$$A_{33} = \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} = 2-0 = 2$$

$$\begin{aligned} \text{Adj. } A &= \begin{pmatrix} -3 & -6 & -3 \\ -3 & -9 & -3 \\ 1 & 4 & 2 \end{pmatrix}^T \\ &= \begin{pmatrix} -3 & -3 & 1 \\ -6 & -9 & 4 \\ -3 & -3 & 2 \end{pmatrix} \end{aligned}$$

$$A^{-1} = \frac{1}{|A|} (\text{Adj. } A)$$

$$= \frac{1}{-3} \begin{pmatrix} -3 & -3 & 1 \\ -6 & -9 & 4 \\ -3 & -3 & 2 \end{pmatrix}$$

Now

$$A \cdot A^{-1} = \begin{pmatrix} 2 & -1 & 1 \\ 0 & 1 & -2 \\ 3 & 0 & -3 \end{pmatrix} \cdot \frac{1}{-3} \begin{pmatrix} -3 & -3 & 1 \\ -6 & -9 & 4 \\ -3 & -3 & 2 \end{pmatrix}$$

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$$= -\frac{1}{3} \begin{pmatrix} -8+6-3 & -6+9-3 & 2-4+2 \\ 0-6+6 & 0-9+6 & 0+4-4 \\ -9+0+9 & -9-0+9 & 3+0-6 \end{pmatrix}$$

$$= -\frac{1}{3} \begin{pmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A^{-1} \cdot A = \frac{1}{-3} \begin{pmatrix} -3 & -3 & 1 \\ -6 & -9 & 4 \\ -3 & -3 & 2 \end{pmatrix} \begin{pmatrix} 2 & -1 & 1 \\ 0 & 1 & -2 \\ 3 & 0 & -3 \end{pmatrix}$$

$$= \frac{1}{-3} \begin{pmatrix} -6-0+3 & 3-3+0 & -3+6-3 \\ -12-0+12 & 6-9+0 & -6+18-12 \\ -6-0+6 & +3-3+0 & -3+6-6 \end{pmatrix}$$

$$= \frac{1}{-3} \begin{pmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Here,

$$A \cdot A^{-1} = A^{-1} \cdot A = I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

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